

AWS Service Consumption Intelligence Platform

Executive Summary

Organizations increasingly rely on AWS services to execute critical business workloads. While AWS provides individual service-level monitoring capabilities, there is often no centralized mechanism to provide consolidated visibility into service consumption patterns, historical trends, stakeholder reporting, and proactive anomaly detection.

This proposal introduces an AWS Service Consumption Intelligence Platform that provides centralized monitoring, automated reporting, AI-powered analysis, and proactive alerting for AWS service utilization across teams and projects.

The platform will operate entirely within the organization's approved infrastructure and security boundaries, utilizing company-approved AI services and maintaining complete data privacy.

Vision

Create a centralized intelligence platform that:

- Monitors AWS service consumption.
 - Provides historical usage analytics.
 - Generates automated reports.
 - Detects abnormal consumption patterns.
 - Sends proactive alerts to stakeholders.
 - Delivers AI-generated insights and recommendations.
 - Operates completely within company-controlled infrastructure.
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Security & Compliance Approach

Internal Deployment Model

The complete solution will be deployed within the company's internal network.

Security Principles

- No public internet access required.
- No external exposure of AWS consumption data.
- Internal authentication and authorization.
- Internal report generation and distribution.

- Internal stakeholder access only.
- Company-approved infrastructure usage.

AI Security

The platform will utilize the organization's approved AI intelligence platform.

Benefits:

- No external AI service dependency.
 - No sensitive operational data shared externally.
 - Compliance with organizational AI governance standards.
 - Alignment with enterprise security requirements.
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Initial AWS Services in Scope

Phase 1 services:

- AWS Glue
- Amazon Redshift
- Amazon S3
- AWS Lambda
- Amazon SES

Additional services can be onboarded in future phases.

Business Objectives

Visibility

Provide centralized visibility into AWS service consumption.

Governance

Enable monitoring of resource usage across teams and projects.

Operational Awareness

Provide early detection of abnormal service consumption.

Reporting Automation

Reduce manual effort in preparing service utilization reports.

AI-Powered Insights

Enable intelligent analysis of service consumption trends.

PHASE 1 – Core Monitoring Foundation

Service Consumption Monitoring

Collect service-level consumption metrics.

AWS Glue

Monitor:

- Job executions
- Successful runs
- Failed runs
- Average execution duration
- Daily usage trends
- Monthly usage trends

Amazon Redshift

Monitor:

- Query executions
- Active sessions
- Query trends
- Cluster utilization metrics

Amazon S3

Monitor:

- Bucket size
- Storage growth
- Object count
- Daily growth patterns

AWS Lambda

Monitor:

- Invocation count
- Execution duration
- Error count
- Success rate

Amazon SES

Monitor:

- Emails sent
 - Bounce rates
 - Complaint rates
 - Delivery statistics
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Historical Trend Analysis

Maintain historical records.

Provide:

- Daily trends
- Weekly trends
- Monthly trends
- Quarterly trends

Examples:

- Service growth trends
 - Team consumption trends
 - Resource utilization trends
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Dashboard

Centralized dashboard providing:

- Service consumption summary
 - Trend charts
 - Historical comparisons
 - Consumption breakdowns
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PHASE 2 – Automated Reporting Engine

Scheduled Email Reports

Configurable reporting cadence.

Daily Reports

Audience:

- Service owners
- Engineering teams

Contents:

- Previous day activity
 - Service utilization summary
 - Failed executions
 - Significant observations
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Weekly Reports

Audience:

- Team leads
- Project managers

Contents:

- Weekly trends
 - Service consumption summary
 - Growth patterns
 - Significant events
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Monthly Reports

Audience:

- Leadership
- Delivery managers
- Technology leads

Contents:

- Monthly summary
 - Trend analysis
 - Usage comparisons
 - AI-generated observations
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Report Formats

- HTML Email
 - PDF Report
 - Dashboard Export
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Report Visualizations

- Trend charts
 - Service comparison charts
 - Historical usage graphs
 - Growth indicators
 - Team-wise comparisons
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PHASE 3 – Alerting & Notification Framework

Consumption Surge Detection

Automatically identify unusual increases.

Examples:

- Glue executions increased 300%
 - Lambda invocations doubled
 - S3 storage growth exceeded historical patterns
 - SES email volume spike detected
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Threshold-Based Alerts

Configurable thresholds.

Examples:

- Daily execution thresholds
 - Storage thresholds
 - Invocation thresholds
 - Email volume thresholds
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Emergency Notifications

Critical alerts sent immediately.

Examples:

- Massive service consumption spike
- Unusual execution behavior
- Potential runaway workloads

Recipients:

- Service owners
 - Project managers
 - Team leads
 - Support teams
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Alert Channels

- Email
 - Internal messaging platforms (future)
 - Dashboard notifications
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PHASE 4 – AI Intelligence Layer

AI Consumption Analysis

Use company-approved AI services to analyze trends.

Examples:

Instead of showing:

"Glue executions increased by 40%"

AI explains:

"Increase primarily driven by Analytics team's ETL workload expansion during the last reporting period."

AI Executive Summaries

Generate stakeholder-friendly summaries.

Examples:

- Weekly summary
 - Monthly summary
 - Leadership summary
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AI Trend Interpretation

Explain:

- Growth trends
 - Seasonal patterns
 - Consumption anomalies
 - Service utilization changes
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AI Recommendation Engine

Provide suggestions such as:

- Review sudden workload growth.
 - Investigate repeated execution failures.
 - Analyze rapidly growing storage usage.
 - Review abnormal email sending patterns.
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Natural Language Querying

Examples:

"Which service had the highest growth this month?"

"Show Lambda usage trend for the last 90 days."

"Which teams generated the most Glue executions?"

"What caused the recent increase in consumption?"

PHASE 5 – Team & Project Intelligence

Team-Level Consumption Monitoring

Track usage by:

- Team
- Project
- Application
- Business Unit

(using available tagging strategies)

Team Comparison Reports

Examples:

- Service utilization by team
 - Growth by team
 - Trend comparisons
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Stakeholder-Specific Reports

Customized reports for:

- Engineers
 - Team Leads
 - Project Managers
 - Delivery Managers
 - Leadership Teams
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PHASE 6 – Advanced Intelligence & Forecasting

(Future Enhancement)

Predictive Consumption Forecasting

Forecast future service consumption.

Examples:

- Next week projections
 - Next month projections
 - Trend-based forecasts
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Capacity Planning Support

Provide estimates for:

- Future growth
 - Service expansion
 - Resource planning
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Forecast-Based Alerts

Notify stakeholders before significant consumption increases occur.

Optional Feature – Cost Intelligence

This feature is intentionally positioned as optional.

Some organizations may prefer not to expose detailed cost information across teams.

If required, the platform can support:

- Service-level cost estimation
- Region-based pricing calculations
- Cost trend monitoring
- Cost forecasting
- Cost optimization recommendations

This feature can be enabled based on organizational approval and governance requirements.

Expected Business Benefits

Improved Visibility

Single platform for AWS service consumption monitoring.

Faster Decision Making

Actionable insights delivered directly to stakeholders.

Reduced Manual Reporting

Automated report generation and distribution.

Early Issue Detection

Proactive identification of abnormal consumption patterns.

Improved Governance

Better operational awareness across projects and teams.

AI-Assisted Operations

Faster understanding of service utilization trends through company-approved AI intelligence.

Success Criteria

- Centralized monitoring of AWS service consumption.
- Automated scheduled reporting.
- Proactive anomaly detection.
- AI-generated operational insights.
- Secure internal deployment.
- Adoption across engineering and delivery teams.

The platform will serve as a foundational AWS Service Consumption Intelligence solution that can be expanded incrementally based on business priorities and organizational requirements.